

❖ Ch4_No.1

The length of the laser gain medium is l , refractive index is η , the length of the laser cavity is L , and the refractive index inside the cavity except the gain medium is η' , the photon density of the gain medium is N , then proof the light with central frequency

$$\frac{dN}{dt} = \Delta n \sigma_{21} c N \frac{l}{L'} - N \frac{\delta c}{L'} \quad 21$$

where $L' = \eta l + \eta' (L - l)$

Proof: $L' = \eta \cdot l + \eta' \cdot (L - l)$

Round-trip: $t_R = \frac{2L'}{c}$

$$\Rightarrow g = \Delta n \cdot \sigma_{21} \rightarrow 2gl \rightarrow 2g$$

$$\Delta N = N(2 \cdot \Delta n \cdot \sigma_{21} \cdot l) - N(2g)$$

$$\Rightarrow \frac{dN}{dt} \approx \frac{\Delta N}{t_R} = \frac{N(2 \cdot \Delta n \cdot \sigma_{21} \cdot l) - N(2g)}{\frac{2L'}{c}}$$

$$\Rightarrow \frac{dN}{dt} = \Delta n \cdot \sigma_{21} \cdot c \cdot N \cdot \frac{l}{L'} - N \cdot \frac{\delta c}{L'}$$

❖ Ch4_No.4

For pulse Nd: YAG laser, the transmittance T_1 and T_2 of the two mirrors are 0 and 0.5 respectively. The diameter of the gain material $d=0.8$ cm, refractive index $\eta=1.836$, total quantum efficiency is 1, fluorescent line-width $\Delta \nu_F=1.95 \times 10^{11}$ Hz, and the spontaneous emission lifetime $\tau_{s2}=2.3 \times 10^{-4}$ s. It is assumed that the average wavelength of the optically pumped absorption band is $\lambda_p=0.8$ μ m. Estimate the threshold pump energy E_{pt} that this laser needs to absorb at the central frequency.

均匀展宽

$$\lambda_0 = 1.064 \mu\text{m}$$

$$g(\nu_0) = \frac{2}{\pi \cdot \Delta \nu_F}$$

$$\Rightarrow \sigma_{21} = \frac{\lambda_0^2}{8\pi \cdot \eta^2 \cdot \tau_{s2}} g(\nu_0)$$

$$\sigma_{21} = \frac{(1.064 \mu\text{m})^2}{4\eta^2 \cdot 1.836^2 \cdot 2.3 \times 10^{-4} \cdot 1.95 \times 10^{11}} \approx 1.897 \times 10^{-22} \text{ m}^2$$

$$\Rightarrow \gamma_{th} = \alpha + \frac{1}{2L} \cdot \ln\left(\frac{1}{R_1 R_2}\right) \leftarrow \alpha = 0$$

$$\gamma_{th} = \frac{\ln^2}{2L}$$

$$\Rightarrow \Delta n_{th} = \frac{\gamma_{th}}{\sigma_{21}} = \frac{\ln^2}{2L \cdot \sigma_{21}}$$

$$V = \frac{\pi d^2 \cdot L}{4}$$

$$\Rightarrow \Delta N_{th} = \Delta n_{th} \cdot V = \frac{\ln^2}{2L \cdot \sigma_{21}} \cdot \frac{\pi d^2 \cdot L}{4} = \frac{\pi d^2 \cdot \ln^2}{8 \sigma_{21}}$$

$$\Rightarrow \Delta N_{th} = \frac{\pi \cdot (0.008)^2 \cdot 0.6931}{8 \cdot 1.897 \cdot 10^{-22}} \approx 9.18 \times 10^{16}$$

$$\Rightarrow E_p = \frac{hc}{\lambda_p} = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{0.8 \times 10^{-6}} = 2.485 \times 10^{-19} \text{ J}$$

$$\Rightarrow E_{pt} = \Delta N_{th} E_p = 9.18 \times 10^{16} \times 2.485 \times 10^{-19} = \underline{\underline{22.8 \text{ mJ}}}$$